

Mohammed Alam

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EDUCATION

Columbia University – *Bachelor of Science in Computer Science*

Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Advanced Programming (C), Database Systems, Web Security, User Interface Design

EXPERIENCE

Columbia Housing

May 2024 – Present

Systems & Operations

New York, NY

- Designed and built internal tools and Node.js automation services supporting operations across 40 residential buildings and 6,000+ rooms, automating workflows and synchronizing Housing data across systems
- Leading development of Dorm Room Viewer after its adoption by Columbia Housing, deploying and maintaining the production platform serving 9,000+ students
- Containerized and deployed Node.js services with Docker on AWS, exposing protected REST APIs aligned with security and data governance requirements set with Directors of Operations and IT Systems

Headstarter AI

July 2024 – September 2024

Software Engineering Fellow

New York, NY

- Shipped 5 production-ready AI applications using Next.js, TypeScript, OpenAI, Pinecone, and Stripe within tight weekly sprints across an 8-week structured fellowship
- Won 1st place among 75+ participants at a TikTok-sponsored hackathon building a real-time blogging platform using React, Node.js, HTML, and CSS with infinite scroll and live updates
- Built RAG pipelines with OpenAI and Pinecone vector search for semantic retrieval across production applications

Freelance

September 2019 – May 2024

Software Engineer

New York, NY

- Delivered 500+ custom web applications for clients using React, Node.js, MongoDB, and Redis
- Developed custom Discord applications for clients, including a multiplayer game that grew to 13,000,000 users across 13,000+ servers

PROJECTS

Mini-Vercel

July 2026 – Present

- Architected a self-hosted deployment platform (TypeScript, Node.js, PostgreSQL, Redis, Docker) that converts GitHub pushes into live HTTPS deployments via HMAC-verified webhooks
- Deployed a self-hosted CI/CD platform on AWS EC2, validating it by migrating personal portfolio off Vercel, and supporting zero-downtime Docker-based releases with health-checked deployments, atomic Nginx routing updates, and instant rollback via previous images
- Wrote a Go reverse proxy for host-based routing with active health checks, plus a CLI and Next.js dashboard controlling the deployment lifecycle (deploy, stop, rollback, custom domains) with live-streamed build logs

Guess The Number

October 2021 – June 2024

- Launched multiplayer Discord bot growing organically to 13,000,000 users across 13,000+ servers
- Reduced command latency by architecting a Redis and MongoDB data layer for high-frequency game sessions
- Processed asynchronous game events with Bull.js queues to stay responsive during concurrency spikes

Dorm Room Viewer | Columbia Housing

June 2025 – Present

- Independently designed and built Dorm Room Viewer, a React housing platform later adopted by Columbia Housing for official development
- Engineered room-level floor plans, walk-time estimation, and filters backed by an AWS S3 image pipeline
- Integrated Columbia CAS (SSO) authentication via NextAuth.js and an admin panel for content approvals

Dine U | Columbia Dining

May 2025

- Created a cross-platform mobile app using React Native, Supabase, and Node.js with real-time menu updates
- Trained ML recommendation system on user dining preferences to surface personalized meal suggestions
- Deployed serverless APIs via Supabase Edge Functions, reducing infrastructure overhead and response times

SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL

Technologies: React, Next.js, React Native, Node.js, PostgreSQL, MongoDB, Redis, AWS, Docker, Git, REST APIs